Numerical Analysis (MAC F241)

Assignment

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Problem 1

Problem Statement

Write a MATLAB code for piecewise linear Lagrange approximation. Compute the approximation to the following functions accurate to within ε.

- $f(x) = \frac{1}{1+x^2}$ in $[-5, 5]$; $\varepsilon = 0.000005$
- $f(x) = x^{\frac{1}{2}}$ in $[0, 1]$; $\varepsilon = 0.00005$

Show $f(x)$ and its approximation graphically and find the computational time.

Formulas For Piecewise Linear Interpolant

Suppose in the given interval we choose an arbitrary partition consisting of $n+1$ points $a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b$. The piecewise linear interpolant of a function $f(x)$ relative to this partition is a function $s(x)$ that satisfies :

1. $s(x)$ is continuous on $[a, b]$.
2. On each subinterval $[x_i, x_{i+1}]$, $i = 0, 1, \cdots, n-1$, $s(x)$ coincides with the linear polynomial

$$s(x) = s_i(x) = a_i + b_i(x - x_i)$$

3. $s(x)$ agrees with $f(x)$ at $x_0, x_1, \cdots, x_n$.

The third condition implies

$$f(x_i) = s(x_i) = s_i(x_i) = a_i \text{ for } i = 0, 1, \cdots, n-1.$$

Hence,

$$a_i = f(x_i) \text{ for } i = 0, 1, \cdots, n-1.$$

To determine b_i's we use the continuity condition,

$$s_i(x_{i+1}) = s_{i+1}(x_{i+1}) \text{ for } i = 0, 1, \cdots, n-1.$$

$$\implies a_i + b_i(x_{i+1} - x_i) = a_{i+1}$$

$$\implies b_i = \frac{a_{i+1} - a_i}{x_{i+1} - x_i} = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}$$

Hence in each subinterval $[x_i, x_{i+1}]$ the coefficients a_i, b_i can simply be calculated by the following formulas :

$$\boxed{a_i = f(x_i)}$$

$$\boxed{b_i = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}}$$

for $i = 0, 1, \cdots, n-1$.

Upper bound for error in Piecewise Linear Interpolation

We are fundamentally partitioning the given interval and on each subinterval $[x_i, x_{i+1}]$, standard linear interpolation is being performed, so the standard linear interpolation error formula gives

$$|f(x) - s_i(x)| = \frac{1}{2}|f''(\xi)||(x - x_i)(x - x_{i+1})|$$

for some ξ such that $x_i < \xi < x_{i+1}$.

Therefore,

$$\max_{x\in[x_i,x_{i+1}]} |f(x) - s_i(x)| \leq \frac{1}{2} \max_{x\in[x_i,x_{i+1}]} |f''(x)| \cdot \max_{x\in[x_i,x_{i+1}]} |(x - x_i)(x - x_{i+1})|.$$

Let $g(x) = |(x - x_i)(x - x_{i+1})|$. On $[x_i, x_{i+1}]$, g attains its maximum value

$$\frac{h_i^2}{4}$$

at $x = \frac{x_i+x_{i+1}}{2}$, where $h_i = x_{i+1} - x_i$.

Substituting this value into the above bound gives

$$\max_{x\in[x_i,x_{i+1}]} |f(x) - s_i(x)| \leq \frac{1}{2} \max_{x\in[x_i,x_{i+1}]} |f''(x)| \cdot \frac{h_i^2}{4} = \frac{1}{8}h_i^2 \max_{x\in[x_i,x_{i+1}]} |f''(x)|.$$

Since

$$\max_{x\in[a,b]} |f(x) - s(x)| = \max_{0\leq i\leq n-1} \left(\max_{x\in[x_i,x_{i+1}]} |f(x) - s_i(x)| \right),$$

it follows that

$$\max_{x\in[a,b]} |f(x) - s(x)| \leq \frac{1}{8} \max_{0\leq i\leq n-1} h_i^2 \cdot \max_{x\in[a,b]} |f''(x)|.$$

Let

$$h = \max_{0\leq i\leq n-1} h_i.$$

Hence,

$$\boxed{\max_{x\in[a,b]} |f(x) - s(x)| \leq \frac{1}{8}h^2 \max_{x\in[a,b]} |f''(x)|}$$

Using this result we will find the a partition of the interval which gives the required precision (say ε). Suppose, for the sake of convenience, we consider equally spaced $n + 1$ points in the interval $[a, b]$, then

$$h = \frac{b - a}{n}$$

By substituting this into the obtained error bound we get (where e is the error),

$$|e| \leq \frac{1}{8}h^2 \max_{x\in[a,b]} |f''(x)| = \frac{(b - a)^2}{8n^2} \max_{x\in[a,b]} |f''(x)| < \varepsilon$$

Hence the value of n can be determined by,

$$\boxed{n > \frac{(b-a)}{2}\sqrt{\frac{1}{2\varepsilon}\max_{x\in[a,b]}|f''(x)|}}$$

Computation of Required Number of Subintervals

Let us find the values of n for the functions given in the problem statement.

- We are given $f(x) = \frac{1}{1+x^2}$ in $[-5, 5]$ with $\varepsilon = 5 \times 10^{-6}$. We compute

$$f'(x) = \frac{-2x}{(1+x^2)^2}$$

and

$$f''(x) = \frac{6x^2 - 2}{(1+x^2)^3}$$

In the interval $[-5, 5]$, the maximum value of $|f''(x)|$ is attained at $x = 0$, which is 2. (Obtained Graphically)

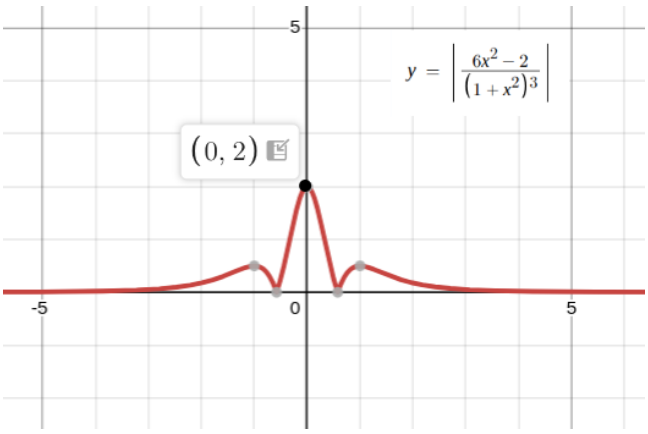


Figure 1: Graph of $y = |f''(x)| = \left|\frac{6x^2-2}{(1+x^2)^3}\right|$

Hence $\max_{x\in[a,b]}|f''(x)| = 2$, $a = -5$, $b = 5$ and $\varepsilon = 5 \times 10^{-6}$. Substituting these values in the derived expression we get

$$n > \frac{(5-(-5))}{2}\sqrt{\frac{1}{2\times 5\times 10^{-6}} \times 2} \approx 2236.067977$$

Hence for $f(x) = \frac{1}{1+x^2}$ we take,

$$\boxed{n = 2237}$$

- Similarly, for $f(x) = x^{\frac{1}{2}}$ in $[0, 1]$ with $\varepsilon = 5 \times 10^{-5}$. We compute

$$f'(x) = \frac{1}{2\sqrt{x}}$$

and

$$f''(x) = \frac{-1}{4} x^{\frac{-3}{2}}$$

which is unbounded as $x \to 0$.

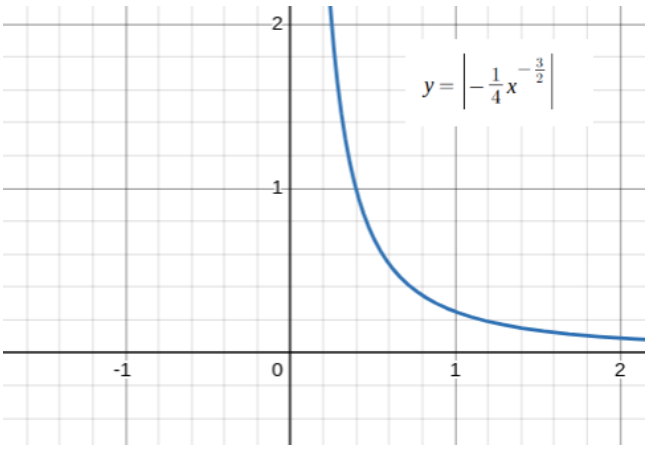


Figure 2: Graph of $y = |f''(x)| = \left| \frac{-1}{4} x^{\frac{-3}{2}} \right|$

Hence,

$$\max_{x \in [0,1]} |f''(x)| = \infty.$$

Therefore, the standard error bound for piecewise linear interpolation is not directly applicable on $[0, 1]$.

To overcome this, we may:

- Avoid including $x = 0$ in the interval, or
- Use a non-uniform partition with finer spacing near $x = 0$.

Since $f''(x)$ is unbounded at $x = 0$, we instead consider a modified interval $[a, 1]$ where a is very close to 0. Let us take $a = 10^{-4}$.

Then,

$$|f''(x)| = \frac{1}{4x^{3/2}}.$$

On $[10^{-4}, 1]$, the maximum value of $|f''(x)|$ is attained at $x = 10^{-4}$:

$$\max_{x \in [10^{-4}, 1]} |f''(x)| = \frac{1}{4(10^{-4})^{3/2}} = \frac{1}{4 \cdot 10^{-6}} = \frac{1}{4 \times 10^{-6}} = 2.5 \times 10^5.$$

Here, $a = 10^{-4}$, $b = 1$, and $\varepsilon = 5 \times 10^{-5}$.

Using the error bound,

$$n > \frac{(b-a)}{2} \sqrt{\frac{1}{2\varepsilon} \max |f''(x)|}.$$

Substituting the values,

$$n > \frac{(1-10^{-4})}{2}\sqrt{\frac{1}{2\times 5\times 10^{-5}}\times 2.5\times 10^{5}}.$$

$$n > \frac{0.9999}{2}\sqrt{\frac{2.5\times 10^{5}}{10^{-4}}} = 0.49995\times\sqrt{2.5\times 10^{9}}.$$

$$n > 0.49995\times 50000 \approx 24997.5.$$

Hence for $f(x) = x^{\frac{1}{2}}$ we take,

$$\boxed{n = 24998}$$

Note that this value of n only guarantees required precision in $[0.0001, 1]$. In $[0, 0.0001)$, the error may be arbitrarily large because $f''(x)$ is unbounded.

MATLAB code for Piecewise Linear Interpolation

We use the following function to build and plot **Piecewise Linear Approximations** :

```
function [] = Problem_1(f, a, b, n)

    tic;
    h = (b - a) / n; % Step Size

    % Choosing Uniformly Spaces n+1 points in the interval [a,b]
    x = linspace(a, b, n+1);

    % Calculate functional values at these points = ai's
    ai = f(x);

    % Initializing array for bi's
    bi = zeros(1, n);

    % Calculating bi's
    for i = 1 : n
        bi(i) = (ai(i+1) - ai(i)) / h;
    end

    toc;

    % Plotting
    figure;
    hold on;
```

```

    % Choosing Dense points (say 10000)
    test_points = linspace(a, b, 10000);
    functional_value = f(test_points);

    % Approx values
    approx_value = zeros(size(test_points));

    for j = 1:length(test_points)

        % Finding the subinterval
        i = floor((test_points(j) - a)/h) + 1;
        if i > n
            i = n;
        end

        % Piecewise linear formula
        approx_value(j) = ai(i) + bi(i)*(test_points(j) - x(i));
    end

    % Plot comparison
    plot(test_points, functional_value, 'b', 'LineWidth', 2);
    plot(test_points, approx_value, 'r--', 'LineWidth', 2);

    legend('Actual f(x)', 'Piecewise Linear Approx');
    title('Piecewise Linear Interpolation');
    xlabel('x'); ylabel('y');
    grid on;

    % Error
    max_error = max(abs(functional_value - approx_value));
    disp(['Maximum Error = ', num2str(max_error)]);

    % Error plot (pointwise error)
    error_values = abs(functional_value - approx_value);

    figure;
    plot(test_points, error_values, 'k', 'LineWidth', 2);

    title('Pointwise Error |f(x) - s(x)|');
    xlabel('x'); ylabel('Error');
    grid on;

```

```
    % Plotting Relative error
    rel_error = (error_values ./ max(abs(functional_value), 1e-12))
        *100;

    figure;
    plot(test_points, rel_error, 'r', 'LineWidth', 2);

    title('Relative Error |f(x) - s(x)| / |f(x)|');
    xlabel('x'); ylabel('Relative Error');
    grid on;

end
```

Function call parameters for $f(x) = \frac{1}{(1+x^2)}$ are as follows :

```
octave:1> f = @(x) 1./(1+x.^2);
octave:2> a = -5;
octave:3> b = 5;
octave:4> n = 2237;
octave:5> Problem_1(f, a, b, n);
```

Function call parameters for $f(x) = x^{\frac{1}{2}}$ are as follows :

```
octave:1> f = @(x) sqrt(x);
octave:2> a = 0;
octave:3> b = 1;
octave:4> n = 24998;
octave:5> Problem_1(f, a, b, n);
```

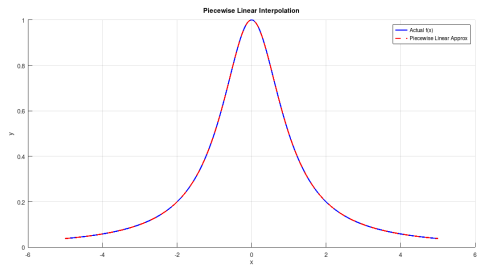
Results for $f(x) = \frac{1}{(1+x^2)}$

For $f(x) = \frac{1}{(1+x^2)}$, the following output is produced :

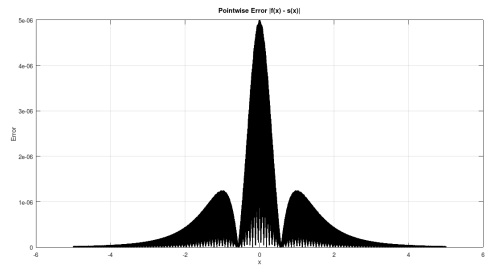
```
Elapsed time is 0.0243161 seconds.
Maximum Error = 4.9943e-06
```

Note that the elapsed time does not include the time taken to generate the plot.

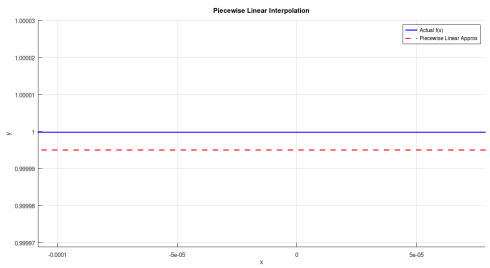
Some Plots



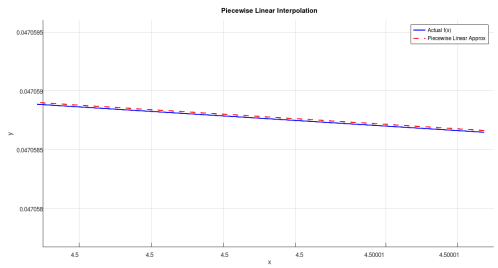
(a) Overall plot in the interval $[-5, 5]$



(b) Point wise Error Plot



(c) Approximation around 0



(d) Approximation around 4.5

Figure 3: Generated Plots

The quality of the approximation is assessed using the relative error. The table below presents the relative error at selected discrete points, while the accompanying plot shows its behavior over the entire interval. Together, these provide a clear view of the accuracy and precision of the approximation.

x	Actual Value	Approx Value	Absolute Error	Relative Error (%)
-1	0.5	0.5000008	7.993294×10^{-7}	0.00016
-0.5	0.8	0.7999994	5.781801×10^{-7}	0.000072
-0.25	0.9411765	0.9411732	3.305552×10^{-6}	0.000351
-0.125	0.9846154	0.9846109	4.518893×10^{-6}	0.000459
0	1	0.999995	4.99581×10^{-6}	0.0005
0.125	0.9846154	0.9846109	4.518893×10^{-6}	0.000459
0.25	0.9411765	0.9411732	3.305552×10^{-6}	0.000351
0.5	0.8	0.7999994	5.781801×10^{-7}	0.000072
1	0.5	0.5000008	7.993294×10^{-7}	0.00016

Table 1: Error analysis for approximation of $f(x) = \frac{1}{(1+x^2)}$

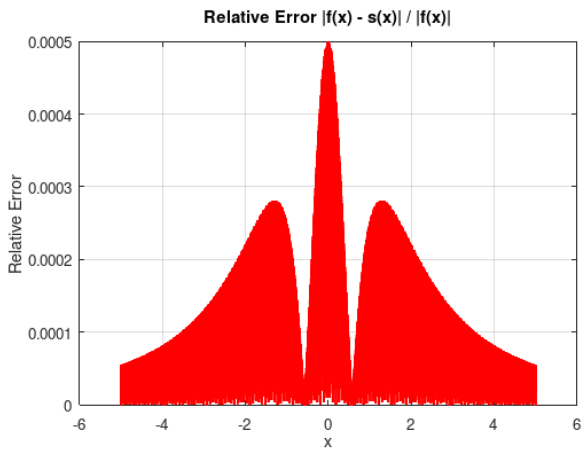


Figure 4: Relative Error Plot for Approximation of $f(x) = \frac{1}{(1+x^2)}$

Observations

1. Since $f''(x)$ is bounded in the interval $[-5, 5]$, overall we get a good approximation of the function. However we had to use a large $n\,(= 2237)$

2. The error in the neighborhood of 0 (i.e, towards the center) is larger than the error in the neighborhood of say 4.5 (i.e, towards the end). This effect can be seen in the point-wise error plot.

3. The observed error behavior is consistent with the theoretical error bound

$$|f(x) - s(x)| \leq \frac{1}{8} h^2 \max |f''(x)|.$$

   Since $|f''(x)|$ is larger near $x = 0$, the error is correspondingly larger in that region.

4. The piecewise linear interpolant captures the general shape of the function well, but fails to accurately approximate regions with higher curvature unless sufficiently small subintervals are used.

5. The use of a uniform partition is not optimal for this function, since the curvature varies across the interval. A non-uniform partition with finer partition near $x = 0$ would yield better accuracy with fewer subintervals.

6. The maximum error obtained numerically is within the prescribed tolerance ε, validating the theoretical estimate used to determine n.

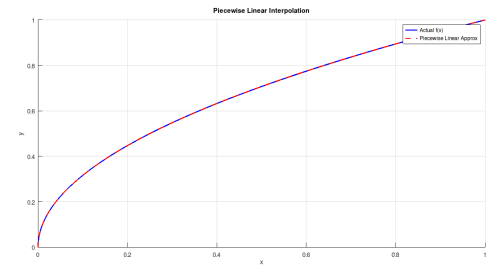
Results for $f(x) = x^{\frac{1}{2}}$

For $f(x) = x^{\frac{1}{2}}$, the following output is produced :

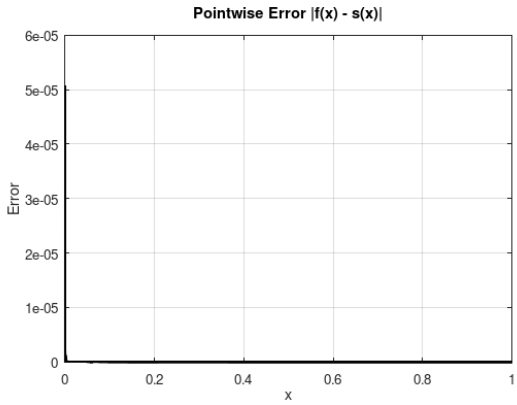
```
Elapsed time is 0.312404 seconds.
Maximum Error = 5.064e-05
```

Note that the elapsed time does not include the time taken to generate the plot.

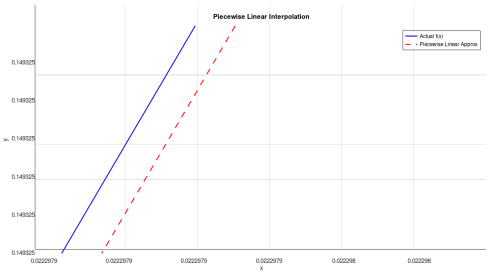
Some Plots



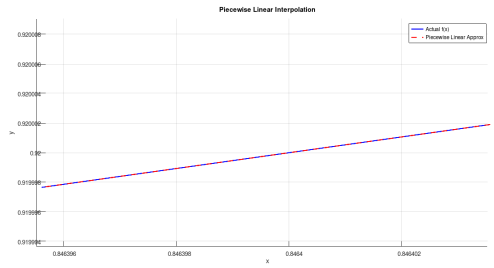
(a) Overall plot in the interval $[0.0001, 1)$



(b) Point wise Error Plot



(c) Approximation close to 0



(d) Approximation close to 1

Figure 5: Generated Plots

The quality of the approximation is assessed using the relative error. The table below presents the relative error at selected discrete points, while the accompanying plot shows its behavior over the entire interval. Together, these provide a clear view of the accuracy and precision of the approximation.

x	Actual Value	Approx Value	Absolute Error	Relative Error (%)
1	1	1	0	0
10^{-1}	3.162278×10^{-1}	3.162278×10^{-1}	1.012212×10^{-9}	0
10^{-2}	1×10^{-1}	1×10^{-1}	3.928174×10^{-9}	0.000004
10^{-3}	3.162278×10^{-2}	3.162276×10^{-2}	1.288382×10^{-8}	0.000041
10^{-4}	1×10^{-2}	9.949357×10^{-3}	5.064252×10^{-5}	0.506425
10^{-5}	3.162278×10^{-3}	1.581076×10^{-3}	1.581202×10^{-3}	50.002
10^{-6}	1×10^{-3}	1.581076×10^{-4}	8.418924×10^{-4}	84.189244
10^{-7}	3.162278×10^{-4}	1.581076×10^{-5}	3.00417×10^{-4}	95.0002
10^{-8}	1×10^{-4}	1.581076×10^{-6}	9.841892×10^{-5}	98.418924
10^{-9}	3.162278×10^{-5}	1.581076×10^{-7}	3.146467×10^{-5}	99.50002
10^{-10}	1×10^{-5}	1.581076×10^{-8}	9.984189×10^{-6}	99.841892

Table 2: Error analysis for approximation of $f(x) = x^{\frac{1}{2}}$

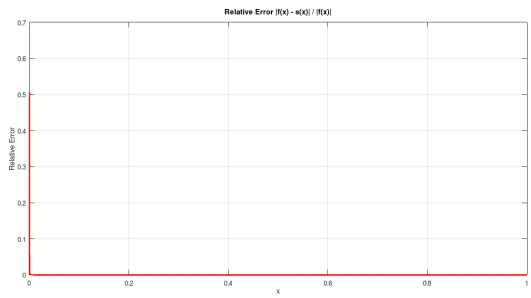


Figure 6: Relative Error Plot for Approximation of $f(x) = x^{\frac{1}{2}}$

Observations

1. Away from 0 we get a good approximation of the function. However we had to use a very large $n\,(= 24998)$
2. Since $f''(x)$ is unbounded around 0, there is a sharp rise in error around it. This effect can be seen in the point-wise error plot and the table above.
3. The maximum error obtained in $[0.0001, 0]$ is numerically close to the prescribed tolerance $\varepsilon = 5 \times 10^{-5}$, which validates the theoretical error estimate used to determine n.
4. The error decreases as we move away from $x = 0$, since the magnitude of $|f''(x)| = \frac{1}{4x^{3/2}}$ decreases with increasing x.
5. The function exhibits very high curvature near $x = 0$, which explains the need for a large number of subintervals in order to maintain the desired accuracy.
6. The use of a uniform partition is inefficient for this function, as most of the subintervals are unnecessarily small in regions where the function is smooth.
7. A non-uniform partition with finer spacing near $x = 0$ and coarser spacing away from it would significantly reduce the required number of subintervals while maintaining the same accuracy.
8. The computational time is higher compared to the previous function due to the significantly larger value of n.